

Lab 5: NoSQL databases

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1 Objectives

The purpose of this lab is to discover two NoSQL databases and manage data with them through two scenarios.

2 Setup

For both scenarios, you don't need to install anything on your computer. Everything will be run online via a sandbox.

3 Presentation

The presentation for Lab 5 is done orally. **All group members** present to the lab assistant which then determines if the demonstrated knowledge is sufficient for a pass. If not, the students can present at a later occasion, same or other day. The lab is graded individually, meaning that one team member can pass while the other fails.

4 Instructions

You must work in a **group of 2 students**, together, but with equal work distribution. Your team will be the same for all the remaining laboratory assignments as well. You should be with a partner that is in the same lab session as you (easier for everybody).

5 Scenario 1:

We will use the online graph database neo4j (Figure 1) available there: <https://neo4j.com/sandbox/>.

Please follow the instructions to run the example database Movies. You **MUST** connect with an account, with Gmail if you have one or please create a new account.

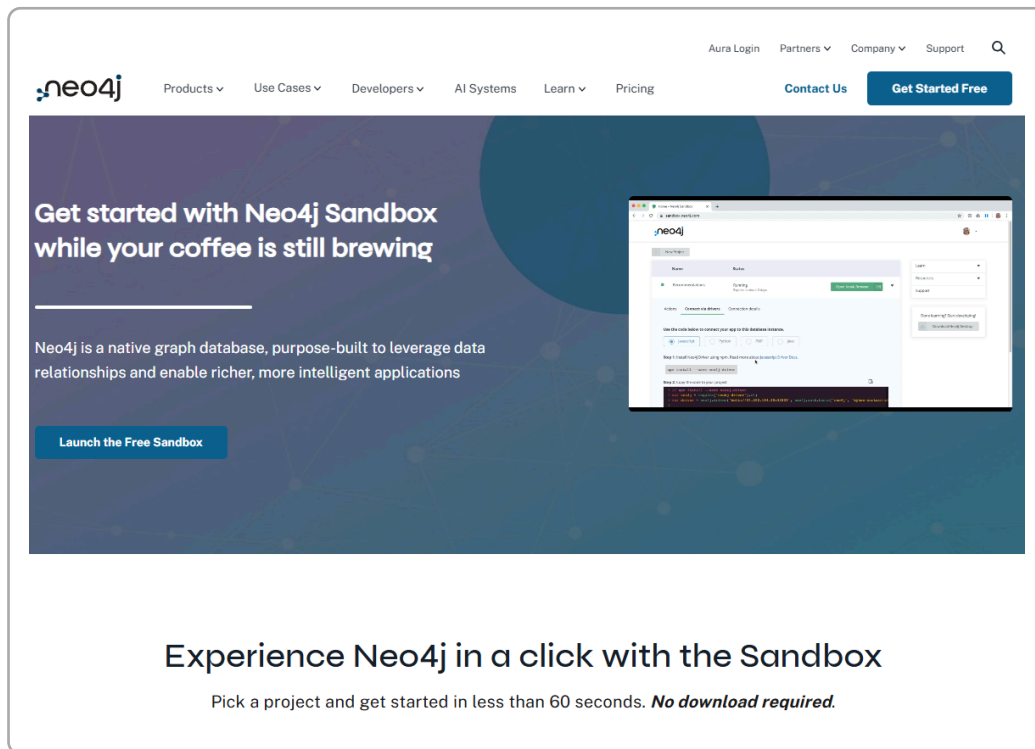


Figure 1: Neo4j Sandbox

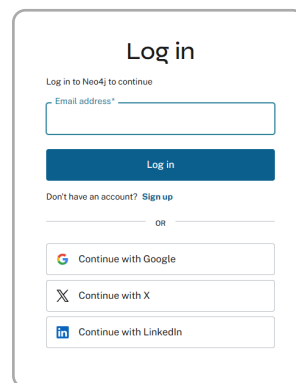


Figure 2: Neo4j Sandbox

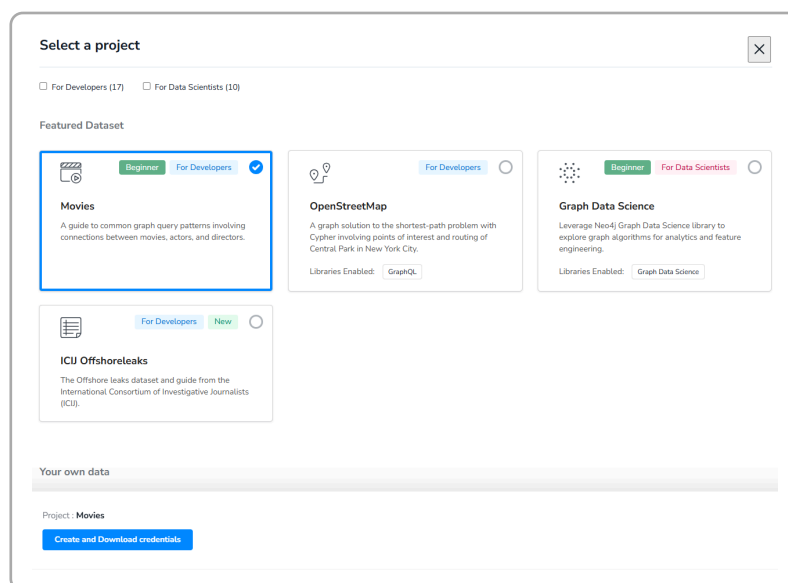


Figure 3: Neo4j Sandbox

- Click on "Create and Download credentials".

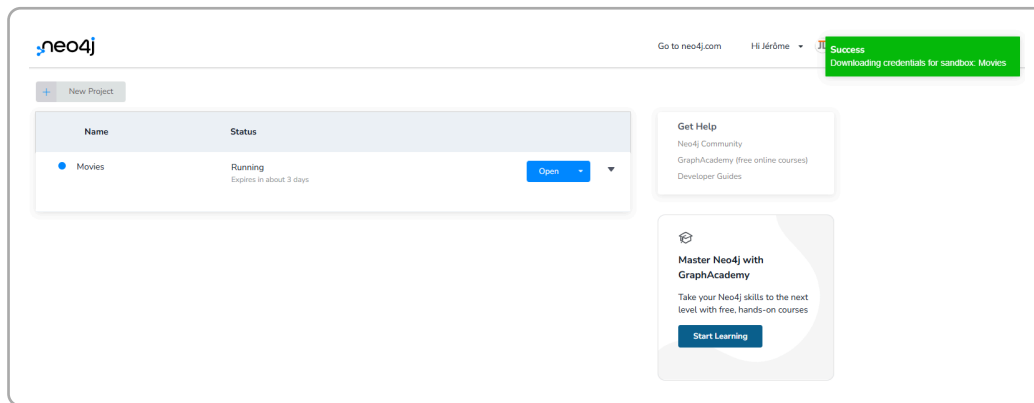


Figure 4: Neo4j Sandbox

- Click on "Open".

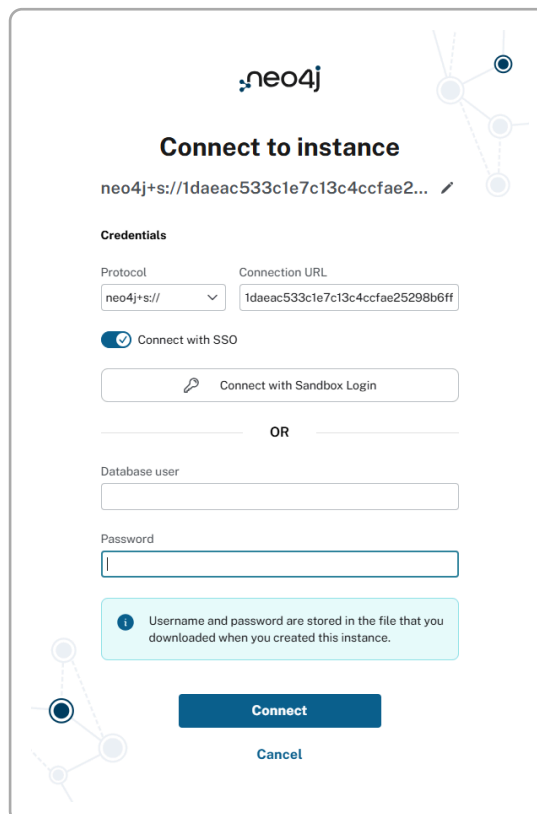


Figure 5: Neo4j Sandbox

- Click on "Connect with Sandbox Login". Please use your account (gmail or other) to connect.

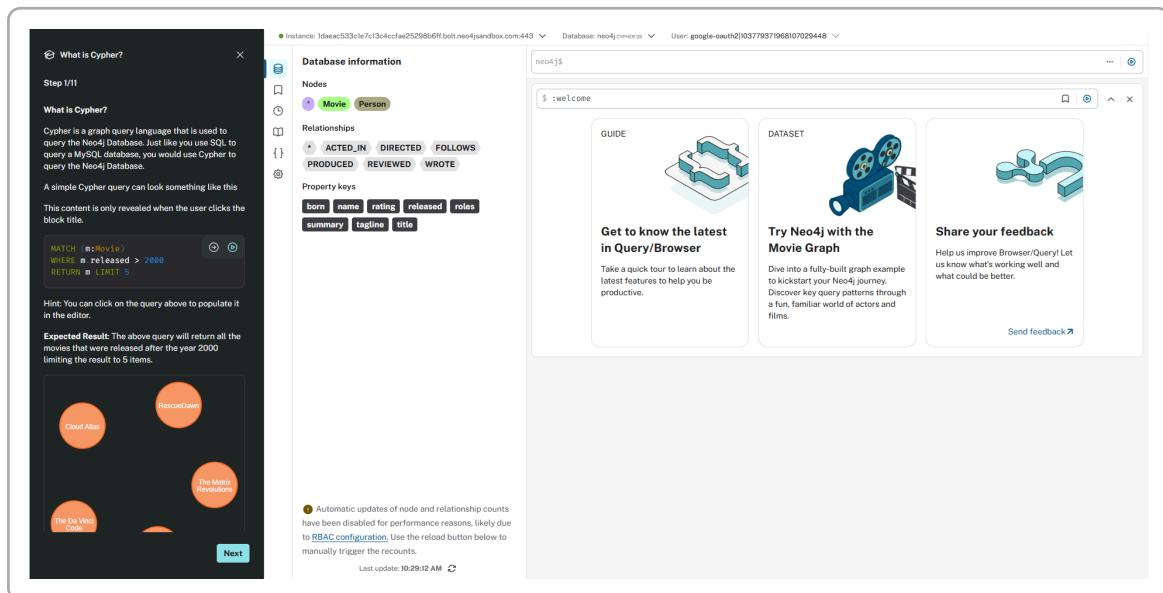


Figure 6: Neo4j Sandbox

- Please run, study and understand all the examples on the left par. There are 11 pages of examples, please **DO ALL OF THEM** to understand the Cypher language syntax.

If you need to reopen your Neo4j sandbox, login to <https://sandbox.neo4j.com/?usecase=movies>.

Questions

1. What is the instruction similar to "SELECT" in neo4j?
2. What is the code to display ALL the database content on the screen? How many nodes are there? How many movies are there? How many persons are there? Try it and play with the graph: zoom in and out, move nodes around...
3. What is the code to create a new Movie (theDatabases, whose title is "The Databases", released in 2026) in this database? Write and test your code on the command line.
4. How do you create a new Person (JeromeL, whose name is "Jérôme Landré" born in 1972) in this database? Write and test your code on the command line.
5. How do you create a new Person (JasminJ, whose name is "Jasmin Jakupovic" born in 1989) in this database? Write and test your code on the command line.
6. How do you make JasminJ play the role of the "lecturer 1" in the movie "The Databases"? Write and test your code on the command line.
7. How do you make JeromeL play the role of the "lecturer 2" in the movie "The Databases"? Write and test your code on the command line.
8. How do you find all actors of the film "The Databases"? Write and test your code on the command line, check that your two new actors are there!
9. Display all the actors of the movie "The Databases" as a list.

6 Scenario 2:

In this scenario, we will use a simplified version of MongoDB online at: <https://mongoplayground.net/>. On the left side of the screen, copy and paste the content (Figure 7) of the file "2026-persons.json" on Canvas:

```
[
  {
    "_id": "69f84a072566e610264a282f",
    "index": 0,
    "favoritepicture": "https://picsum.photos/200",
    "age": 34,
    "fname": "Herring",
    "lname": "Edwards",
    "gender": "male",
    "company": "STEELFAB",
    "email": "herringedwards@steelfab.com",
    "phone": "+1 (868) 522-3917"
  },
  ... MANY MORE PERSONS ...
  {
    "_id": "69f84a07b774e8b3d10e067b",
    "index": 99,
    "favoritepicture": "https://picsum.photos/200",
    "age": 47,
    "fname": "Decker",
    "lname": "Sheppard",
    "gender": "male",
    "company": "TOURMANIA",
    "email": "deckersheppard@tourmania.com",
    "phone": "+1 (905) 595-3469"
  }
]
```

Be careful, the example above is not **complete**, there are more people in the JSON file!

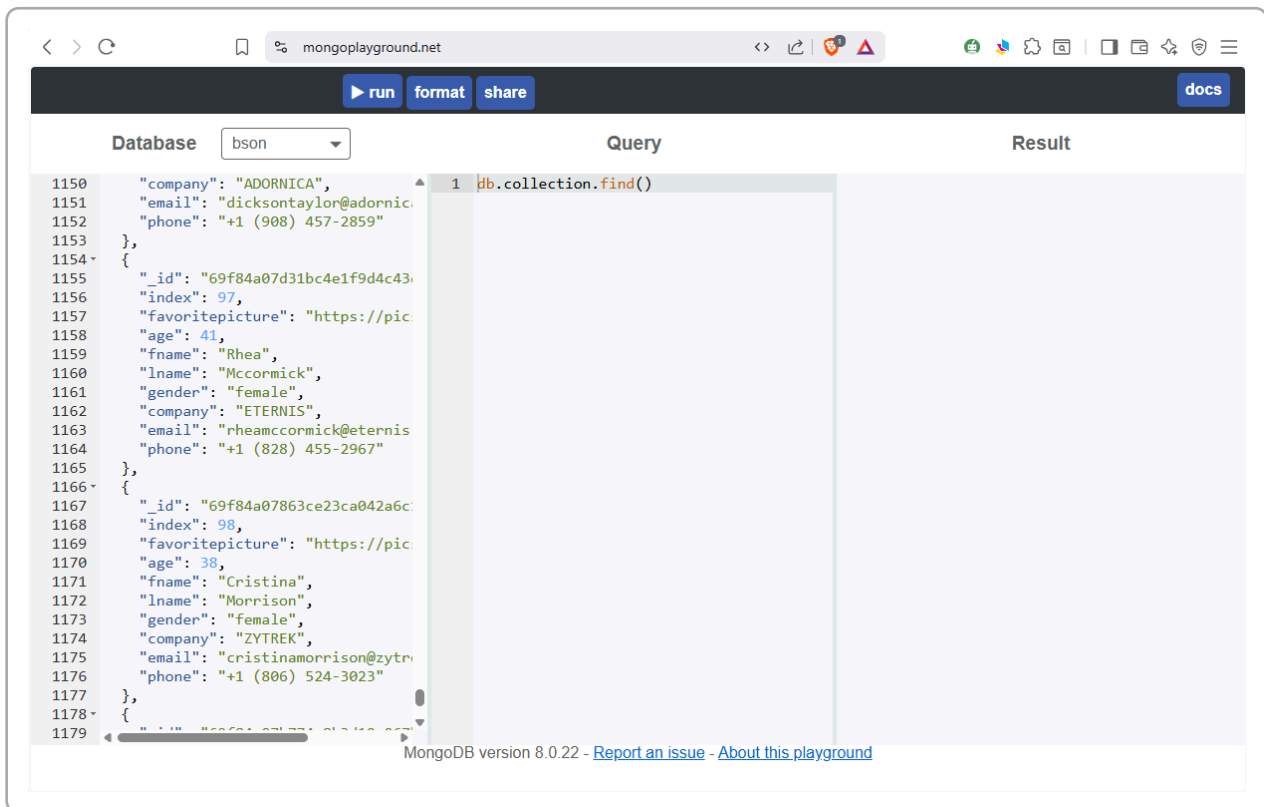


Figure 7: MongoDB playground online.

This will be our starting “collection” of data. A “collection” in MongoDB is similar to a table in SQL.

Please answer, the questions below:

- 1) What is the type of data representation used in MongoDB?
- 2) Do all the objects have the same properties?
- 3) What is the query to display all the data? Type it in the “Query” textbox and run it.
- 4) What is the query to count how many persons there are in the collection? Type it in the “Query” textbox and run it.
- 5) What is the query to display all the persons in ascending alphabetical order of the first name? Type it in the “Query” textbox and run it.
- 6) What is the query to display all the persons in descending alphabetical order of the first name? Type it in the “Query” textbox and run it.
- 7) What is the query to display all the persons in ascending alphabetical order of the last name? Type it in the “Query” textbox and run it.
- 8) What is the query to display all the persons having age greater or equal to 50? Type it in the “Query” textbox and run it.
- 9) What is the query to count the persons having an age lower than 22? Type it in the “Query” textbox and run it.
- 10) What is the query to display all the persons who work for the company “ISODRIVE” (this company have three sites, A, B and C)? Type it in the “Query” textbox and run it.
- 11) What is the query to count all the persons who work for the company “ISODRIVE” (this company have three sites, A, B and C)? Type it in the “Query” textbox and run it.

- 12) What is the query to display the min, max and average age of all the persons? Type it in the "Query" textbox and run it.
- 13) What is the query to display how many persons are female? Type it in the "Query" textbox and run it.
- 14) What is the query to display the fields _id, firstname, lastname, company of "Jérôme Landré"? Type it in the "Query" textbox and run it.
- 15) What is the query to display all the persons working for a company whose name contains "INA"? Type it in the "Query" textbox and run it.
- 16) What is the query to display all the companies (and only the company name) whose name contains "INA"? Type it in the "Query" textbox and run it.